

Samanpreet Kaur

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SKILLS

Programming: C++, Java, Python, JavaScript, HTML, CSS, SQL, React, ExpressJS, Kotlin, DSA, Cloud Computing

Database: DynamoDB, SQL (MS SQL Server), PostgreSQL, Firestore

Tools: AWS (EC2, S3, VPC, IAM), Git, IntelliJ IDEA, Visual Studio Code, Android Studio, MATLAB, Microsoft Office skills

Soft Skills: Teamwork, problem-solving, eye for detail, confident, leadership, strong communication, public speaking

PROFESSIONAL EXPERIENCE

Teaching Assistant, Simon Fraser University, Burnaby, BC | Coding, research, technical writing

August 2025 -current

Data Structures and Programming; Social Issues & Communication Strategies in Computing Science

- Facilitate weekly one-on-one office hours to provide instructional support, exam grading, and constructive feedback, resulting in measurable improvement in student writing and technical communication skills.
- Translate complex technical concepts into clear, student-friendly explanations, helping solve coding problems, fostering academic confidence which help them deepen DSA knowledge and frame their essays well.
- Collaborate with faculty and fellow TAs to coordinate instruction, manage grading timelines, and demonstrating effective teamwork which ensures delivery of necessary feedback timely.

CODING PROJECTS

Ferry Reservation System | C++, GitHub, Binary File I/O, Requirement Analysis, Kanban

May 2025- August 2025

- Developed a **C++ application** to manage ferry reservations, vehicle check-ins by applying **agile practice**, closely following Software Development Life Cycle (**SDLC**) and real-world system design.
- Prepared all **software engineering documents** from scratch including requirements analysis, UI, architectural and detailed design document with deep emphasis on scrutiny, features and design tradeoffs.
- Implemented binary file I/O to handle the data, conducted peer code reviews, performed big-bang integration, and **unit testing** with over 80% statement coverage.
- Coordinated with team using the version-control on **GitHub** and **Kanban system** to maintain consistency which promoted workflow transparency.

Backstage Concert Companion Mobile Application | Kotlin, Android Programming, Firebase, MVVM

October 2025- December 2025

- Developing a **concert-tracking application with an intuitive multi-tab interface**, enabling users to explore events, connect with friends, and manage interests through a smooth, modern UI **using MVVM architecture**.
- Implementing **real-time event search by integrating Ticketmaster's API**, allowing users to browse local and international concerts with live updates and personalized content based on location and preferences.
- Building a scalable user account system **using Firebase Authentication and Firestore**, supporting secure login, profile management, saved events, and synced cross-device data for a consistent user experience.

Mobile Workout Tracking Application | Kotlin, Android Studio, SQLite

September 2025- December 2025

- Designing an Android fitness application with **interactive UI enabling** users to record, view, and manage workouts.
- Integrating 3 workout modes: manual input, GPS-based tracking using Google Maps API, and automatic activity inference leveraging accelerometer sensor data, using **machine learning classifier** and **Kotlin**.
- Built a user profile module with camera/gallery integration for profile images, privacy preferences, and unit selection.
- Optimizing data persistence and retrieval by implementing **SQLite database management** and **Android lifecycle handling** for reliable performance.

Squat Posture Classifier Machine Learning | Python, Machine Learning Algorithms

May 2025- July 2025

- Implemented a KNN-based classifier to detect correct vs. incorrect squat posture using a video input from client using python, scikit-learn.
- Optimized accuracy through feature engineering and model tuning to achieve 80% accuracy.
- Integrated MediaPipe pose estimation to extract key landmarks, enabling reliable real-time feature extraction.

EDUCATION

Bachelor of Computer Science; concentration in artificial intelligence (AI)

Simon Fraser University, Burnaby, BC

January 2025 - **August 2027** (Expected Graduation)

Associate of Computer Science

Langara College

August 2022-December 2024

ADDITIONAL EXPERIENCE

Student Ambassador, Simon Fraser University, Burnaby, BC

January 2026 – current

- Support promotion of university's Faculty of Applied Sciences program by engaging with prospective students through outreach communication and information sessions, helping them understand academic programs and campus opportunities.
- Strengthen the FAS student community by interacting with current students at campus events and providing peer guidance on navigating academic and campus resources via sharing own academic personal journey.

Security Operator, Amazon Web Services (AWS), Vancouver, BC

December 2022 – current

- Communicate construction updates and safety concerns with over 100 visitors and contractors by answering questions and giving building access to create a positive everyday experience.
- Coordinate with cross-functional teams to identify and resolve security issues proactively, contributing to a safe and seamless operational workflow.

Student Engagement Assistant, Langara College, Vancouver, BC

April 2024 -December 2024

- Handled college's inventory for club activities by maintaining Excel sheets and inventory tracking systems to provide resource management and organized student events, clubs and cultural fairs smoothly.
- Promoted awareness about campus resources by answering over 50 students and staff inquiries about wayfinding and help options to make campus life accessible for all.

VOLUNTEER EXPERIENCE

Coding Tutor / Mentor, Women in Computer Science (WiCS), SFU

January 2026– current

- Mentor WiCS students in algorithmic problem solving by teaching LeetCode-style coding problems and guiding them through efficient solution strategies in 1.5-hour study sessions.
- Foster collaborative technical learning sessions by breaking down data structures and algorithm concepts to help peers improve coding interview readiness.

Hack-The-Sem Ambassador, Google Developer Student Club, SFU

January 2026– current

- Promote the Hack-The-Sem innovation program by encouraging student participation and sharing opportunities across personal academic and professional networks.
- Contribute to program visibility and engagement by supporting outreach initiatives that attract students interested in building impactful technical projects.

Vancouver Police Associate, Kitsilano Fairview Community Policing Centre, Vancouver, BC

August 2023 – May 2025

- Resolved 15+ resident phone queries per day by directing callers to appropriate police or city services, leading to residents feel more aware and improve their access to community resources significantly.
- Partnered with Vancouver Traffic Police and ICBC to check the speed of 100+ vehicles per shift using radar equipment, assisting in enforcement actions to halt speeding.

School Captain | Leadership Involvement , PKS International School, India

April 2016 – May 2019

- Led a school House team for 2 consecutive years, organizing and training students for 10+ extracurricular activities including public speaking and games, contributing to winning the *Best House Trophy* both years.
- Served as class monitor for 8 years by supporting teachers in test marking which maintained class discipline.
- Anchored school's Annual Event 4 times by introducing dance items, skits, and announcing prize winners, providing an enjoyable and well-led experience to the audience.